

Model Solution with Bargained Prices

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Surplus sharing in transaction i:

$$p_i = (1-\beta) (1 + \tau(x)) \underline{p}$$

All transactions are same: $p_i = p$ for all.

Surplus sharing becomes:

$$1 = (1-\beta) [1 + \tau(x)]$$

$$\tau(x) = \frac{1}{1-\beta} - 1 = \frac{\beta}{1-\beta}$$

$$x = \tau^{-1}\left(\frac{\beta}{1-\beta}\right)$$

Tightness
under surplus
sharing

We know that x is given by $AD = AS$

$$y^d(x, p) = y^s(x)$$

Price p under surplus sharing is such

$$y^d\left(\tau^{-1}\left(\frac{\beta}{1-\beta}\right), p\right) = y^s\left(\tau^{-1}\left(\frac{\beta}{1-\beta}\right)\right)$$

→ unique p that satisfies this condition

→ aggregate bargained price

$$y^d(x, p) = \frac{X^\varepsilon}{[1 + \tau(x)]^{\varepsilon-1}} \frac{\mu}{p}$$

$$y^d(\tau^{-1}(\beta/(1-\beta)), p) = \frac{x^\varepsilon}{[1 + \beta/(1-\beta)]^{\varepsilon-1}} \cdot \frac{\mu}{p}$$

$$= \frac{x^\varepsilon}{[1/(1-\beta)]^{\varepsilon-1}} \cdot \frac{\mu}{p}$$

$$y^d(\tau^{-1}(\frac{\beta}{1-\beta}), p) = \underline{x^\varepsilon (1-\beta)^{\varepsilon-1}} \cdot \frac{\mu}{p}$$

$$y^s(\tau^{-1}(\beta/(1-\beta))) = \underline{f(\tau^{-1}(\beta/(1-\beta)))} \cdot k$$

Bargained price satisfies:

$$p = \frac{x^\varepsilon (1-\beta)^{\varepsilon-1} \cdot \mu}{f(\tau^{-1}(\beta/(1-\beta))) \cdot k}$$

$$p = \frac{x^\varepsilon \cdot \mu}{k} \times \frac{(1-\beta)^{\varepsilon-1}}{f(\tau^{-1}(\beta/(1-\beta)))}$$

bargaining power

AS

AD

$$p^m = [\quad]$$

price norm capturing surplus sharing.